Cal Hacks 13.0 Master Plan v8: Assisted Evacuation Coordination

Oct 24, 2026 · Brendan Giang · supersedes v7 (Oct 23). Built from three outside reviews of v7 and the team's decisions on them.

TL;DR

We connect an evacuation warning to a confirmed, needs-matched ride for people who can't leave on their own. Wildfire is the first demonstration.

- **The question we answer:** an evacuation warning has gone out. Who still can't leave, and who has actually agreed to pick them up?
- **What we demo: Rider Check.** A paratransit agency loads its rider list, draws the warning zone, and our voice agent calls every rider inside it. Answers become dispatch requests; an optimizer matches each one to a vehicle that can actually do the pickup (space *and* crew); a dispatcher approves; a driver accepts on their own phone; only then does the rider get a call back.
- **The moment judges remember:** one rider whose situation changes. The caller corrects an answer mid-call, the system rejects a van that can't do the pickup, the driver accepts, then a road closes. The stale acceptance is rejected, the rider is told the truth right away, and a new driver is found.
- **Honesty rule:** every update reflects the latest confirmed assignment, and the resident hears right away if it changes. What the agent may say is tied to request status in code, not in the prompt.
- **Proof standard:** we report dispatch gains against a competent heuristic on identical requests, separately from outreach gains; call accuracy per field; and failure cases (cancellations, stale acceptances), not only reassurance pressure.
- **Where it goes next:** ready for a supervised drill with fictional riders, then a shadow-mode trial alongside an agency's own dispatch. County mode (every household, an alert phone number, all vehicle types) is the roadmap, not the demo.
- **Core line:** The vans were ready at 4am. The call came at 6:03. Nothing connected them.
- **Deadlines:** submissions close Oct 25 at 8:50am Pacific. Judging at our table at 10:30am; closing ceremony at 2pm.
- **Open:** the product name.

What changed from v7, and why

CHANGE	WHY
Rider Check is the product and the demo; county mode is one roadmap line	All three reviews landed here. It is the only version that meets Anthropic's "usable on Monday," Microsoft's customer criterion, and Cartesia's "one complete user flow" at once.
Positioning: assisted evacuation coordination, wildfire first	Moves us off ground already covered by Orion and FirstResponder Relay, and names what is new: confirmed, needs-matched pickup.
Commander approval and driver acceptance are separate steps	v7 contradicted itself (page 5 said callbacks wait for the driver; page 12 said approval triggers them). Approval is internal; only acceptance reaches the resident.
Plan versions, stale-acceptance rejection, idempotent callbacks, a `reassigning` status	"Driver accepted" does not mean the promise can be kept. Roads close afterward. This is also the strongest Convex demonstration we have.
Crew and assistance requirements on households and vehicles	A wheelchair space is not proof a vehicle can do the pickup. In the case study, the son needed two or three people to get out of bed.
Corrections are a first-class voice feature; Spanish end to end; per-field accuracy	Cartesia's judges will test recovery, not clean answers. "Any language" is unprovable; one extra language working is provable.
Case study narration fixed	3:25am was the evacuation *order*, and west Altadena got no warning first. Early start is presented as a counterfactual.
Need model becomes a transparent door-check queue with fairness tests	Easier to defend than an unvalidated household posterior; a waiting-time term keeps isolated homes from staying last forever.
Synthetic population: real PUMS records reweighted with IPF	Public PUMS is located only to the PUMA, so v7's block-group conditioning join doesn't work. Resampling real records already keeps the dangerous combinations.
Simulator splits dispatch, outreach, and combined gains	Otherwise the optimizer looks better only because it got earlier requests.
Fire model is a labeled scenario input in the core; the simple spread model returns after the gate as a scenario generator, without a FIRMS validation claim	Satellite detections are too coarse and intermittent to validate address-level arrival times.

CHANGE	WHY
Cut: flow matching and spline flow, volunteer tier, federated-learning close (kept for Q&A), Fetch.ai, Bitdeer, SpaceXAI, Mana, Meta, Moss	Each either makes the project easier to attack or adds a bolt-on a sponsor will penalize. Congestion and drift-diffusion stay, after the gate, as labeled exploration.
"Usable Monday" replaced	"Supervised drill, then shadow mode" is still ambitious and survives a Microsoft question.

The problem

In every major disaster for 20 years, the people who die are disproportionately the ones who can't leave on their own, and nothing reliably connects them to the vehicles that could get them out.

The case study (Eaton Fire, January 2025). A disabled father who used a wheelchair and his son with cerebral palsy lived in west Altadena. Relatives trying to reach them were turned away at the perimeter. The evacuation **order** for west Altadena came at 3:25am; no warning preceded it. Access Services, the county's paratransit provider, said it was ready by 4am and was never called. The father called 911 at 6:03am and again at 6:14; a relative called at 6:35. The dispatcher said "we have a lot of calls in that area." Both men were found dead around 8:30am. The son needed two or three people to get him out of bed.

How we tell this story. Say "a father and son," not their names. Present it as a coordination failure the reporting supports: vehicles ready, nobody called. Do **not** imply our prototype would have saved them, or that reassurance alone explains why they stayed. Our early start is a counterfactual: "What if calls had started as soon as a warning covered their street?"

DISASTER	DEATHS	WHO DIED
LA County fires (2025)	30	Mostly older, many disabled. Average age of the 17 Eaton victims counted then: 77
Lahaina, Maui (2023)	102	More than two-thirds in their 60s or older
Camp Fire, Paradise (2018)	85	A majority over 65
Hurricane Katrina, Louisiana (2005)	971	Almost half 75 or older

- **Can't leave alone:** only 20% of people with disabilities can evacuate immediately without difficulty; 6% can't at all (UN survey, about 6,000 people).
- **Why the system fails:** investigators found evacuations were never formally part of incident command, field reports didn't reach commanders, and the sheriff was short on vehicles.

Dispatchers fall back on reading addresses over the radio. Nothing matches vehicles to people by need.

- **Why registries fail:** during Hurricane Harvey, fewer than 5% of Texas registrants were contacted. LA County shut down its earlier registry for "low return on investment." Advocates warn registries create false assurance.
- **What exists today:** one-way alerts (phone alerts, reverse 911, Watch Duty), zone software (Genasys), keypad polling (Everbridge), inbound AI hotlines. Nothing does outbound conversational triage plus confirmed, needs-matched transport.

Positioning and the opening stat

We are **assisted evacuation coordination for people with access and functional needs (AFN)**, with wildfire as the first demonstration. Emergency managers use "AFN"; using it tells a public-safety judge we did our homework. Hurricanes and floods get one line in the close: the pipeline generalizes, but each hazard needs its own operating rules (days of warning and shelter-in-place calls for hurricanes; high-clearance vehicles for floods). We don't build for them.

> SAY THIS
>
> "Disasters force about 26 million evacuations and displacements a year on average. A warning tells people to leave. It doesn't find out who can't, or confirm that the right vehicle is coming. Only 1 in 5 people with disabilities can evacuate right away without difficulty. We connect the warning to the ride."

- **What the 26 million is:** roughly the ten-year average of IDMC's disaster displacement count. 2025 had 29.9 million, 13% above that average; 2024 had 45.8 million. It counts movements, not people, and most happen in Asia.
- **Never put it next to a fire image.** Wildfires were about 694,000 of the 2025 total.
- **Short on time?** Drop the global figure and lead with the 1-in-5 number and the Eaton Fire. For this audience the specific failure persuades more than the market size.

Related projects (team awareness only, not for the pitch or Devpost)

Two past Berkeley hackathon projects sit near ours, and judges may have seen them. We never mention either first.

- **Orion** (Cal Hacks 12, won Vapi and Warp): a wildfire operations tool for commanders. Fire map, voice commands, routing the nearest station's units to fires.
- **FirstResponder Relay** (UC Berkeley AI Hackathon, June 2026, finalist): triages overflow 911 calls in 13+ languages; Claude extracts urgency and address; a dispatcher map checks callers against fixed perimeters and relays translated replies. Inbound only, and it gives callers a calming reply before any help is assigned.

	ORION	FIRSTRESPONDER RELAY	US
Unit of work	A fire incident	An overflow 911 call	A person who can't get out, matched to a vehicle that can do the pickup
Who it reaches	Nobody; commander-facing	Only people who call in	Every registered rider in the zone, by outbound call; silence goes to door checks
After intake	Nearest station's units to a fire	A ranked list for the dispatcher	A compatible assignment (space and crew), dispatcher approval, driver acceptance on the driver's own phone
When things change	Re-route units	Not addressed	Versioned plans; stale acceptances rejected; resident told right away
What it tells residents	Nothing	A calming reply right away	Nothing promised until a driver accepts; told immediately if that changes

> **ONLY IF A JUDGE BRINGS ONE UP**
>
> "They help responders see the fire or hear the calls that come in. We find the riders who never call, put them on a vehicle that can actually lift them, and only tell them help is coming once a driver has said yes."

Never call our interface a command center or a "dashboard." It's the **rescue map**.

What we're building

One rider's path, from the moment a warning covers their street to the moment they hear who's coming, and what happens when that changes.

1. **Load and target.** The agency uploads its rider list (CSV), with mobility needs already on file. A dispatcher draws or selects the warning zone. Riders inside it enter the call queue, ordered by deadline and known need.
2. **Have a real conversation.** The agent confirms rather than interrogates: are you still at this address, can you leave now, who's with you, any change to mobility or medical needs, how much help do you need getting out (for example "two people to lift"). It fills a structured record during the call and reads it back. English and Spanish, end to end. Before hanging up: "Is there anyone else on your street who might not be able to leave?"
3. **Handle corrections.** "My husband uses a wheelchair, but he isn't here" has to *remove* a wheelchair need already recorded, and the change has to reach the assignment. Corrections are tested, measured, and shown live.

4. **Never treat silence as safe.** No answer means retry at 2 and 5 minutes, then a text, then the door-check queue for field teams.
5. **Plan pickups that can actually happen.** The optimizer matches each request to a vehicle with the right space (seat, wheelchair, stretcher) *and* the right crew (lift-capable, enough people), before the scenario deadline. It shows why a nearby vehicle was rejected.
6. **Approve, then confirm.** The dispatcher approves a plan version on the rescue map. Each affected driver gets an offer on their phone. Approval is internal: the resident hears nothing new until a driver accepts.
7. **Close the loop.** On acceptance, the rider gets a callback by template: "A wheelchair-accessible van with a two-person crew is 6 minutes away. The driver's name is Maria. I'll call again if anything changes."
8. **Tell the truth when it changes.** A road closes or a driver cancels: the assignment is revoked, the rider is told within seconds, the plan re-solves, and any acceptance of the old plan is rejected.

Honesty rules (enforced in code)

The 911 dispatcher's "help is on the way" is part of this story. Our agent cannot say it unless it is true, and must say so when it stops being true.

REQUEST STATUS	WHAT THE AGENT MAY SAY	WHAT IT MAY NEVER SAY
`received`	"You're on the rescue list as high priority. No driver has accepted yet. I'll call you back the moment one does. If you can leave safely, leave now. If fire reaches your home, call 911."	Any promise of pickup or ETA; reassurance in free text ("don't worry," "they'll be there soon")
`approved` (internal)	Same as `received`. Dispatcher approval is not a promise; the resident hears nothing new.	"Someone has been assigned"
`driver_accepted`	Driver name, vehicle type, crew, ETA from the optimizer, inserted by template; "I'll call again if anything changes."	An ETA Claude generated itself
`en_route`	Updated ETA from the vehicle's live position	"Stay put no matter what"
`reassigning`	"Your driver can no longer reach you. You're still at the top of the list. I'll call the moment a new driver accepts. If you can leave safely, leave now. If fire reaches your home, call 911."	The old ETA; silence
`arrived` / `closed`	Confirmation, shelter name	Anything about other residents

- **Order of events, in code and in the diagram:** dispatcher approves plan v*N*, then the driver is offered the assignment, then the driver accepts, then the callback fires. Never approval to callback.

- **Hard escalations:** an address the agent can't confirm after two tries, a medical emergency, or "I can see flames." The agent tells the caller to call 911 if they can, **and** the record goes red at the top of the dispatcher's list, so nobody is just handed back to the line that failed the father and son. The "if you can leave safely, leave now" script goes to disability advocates for review; it has to work for people who can't.
- **Red-team set:** 25 callers who push for reassurance, plus 10 change cases: driver cancels, road closes after acceptance, stale acceptance of an old plan, duplicate events, and a status change while the agent is mid-sentence.
- **Two honesty numbers, not one.** Template-level false promises should be zero by construction, so we also measure what can fail: how often Claude's free text *implies* help is coming while status is `received` or `reassigning`, scored by a separate grader model with a human spot check. Plus: seconds from revocation to the resident being told.

Rider Check now, county mode later

	RIDER CHECK (THIS WEEKEND)	COUNTY MODE (ROADMAP)
Buyer	A paratransit agency like Access Services	County emergency management, with fire and sheriff dispatch
Who gets reached	Registered riders inside the warning zone	Every household: an alert number inbound, registries and landlines outbound
Vehicles	The agency's own vans and crews	Paratransit, transit buses, fire units, ambulances
What it needs to start	The rider list (CSV) and a zone boundary	An alerting authority adding our number to its alert
Next real step	A supervised drill with fictional riders, then shadow mode alongside the agency's dispatchers	One emergency operations center, after Rider Check works

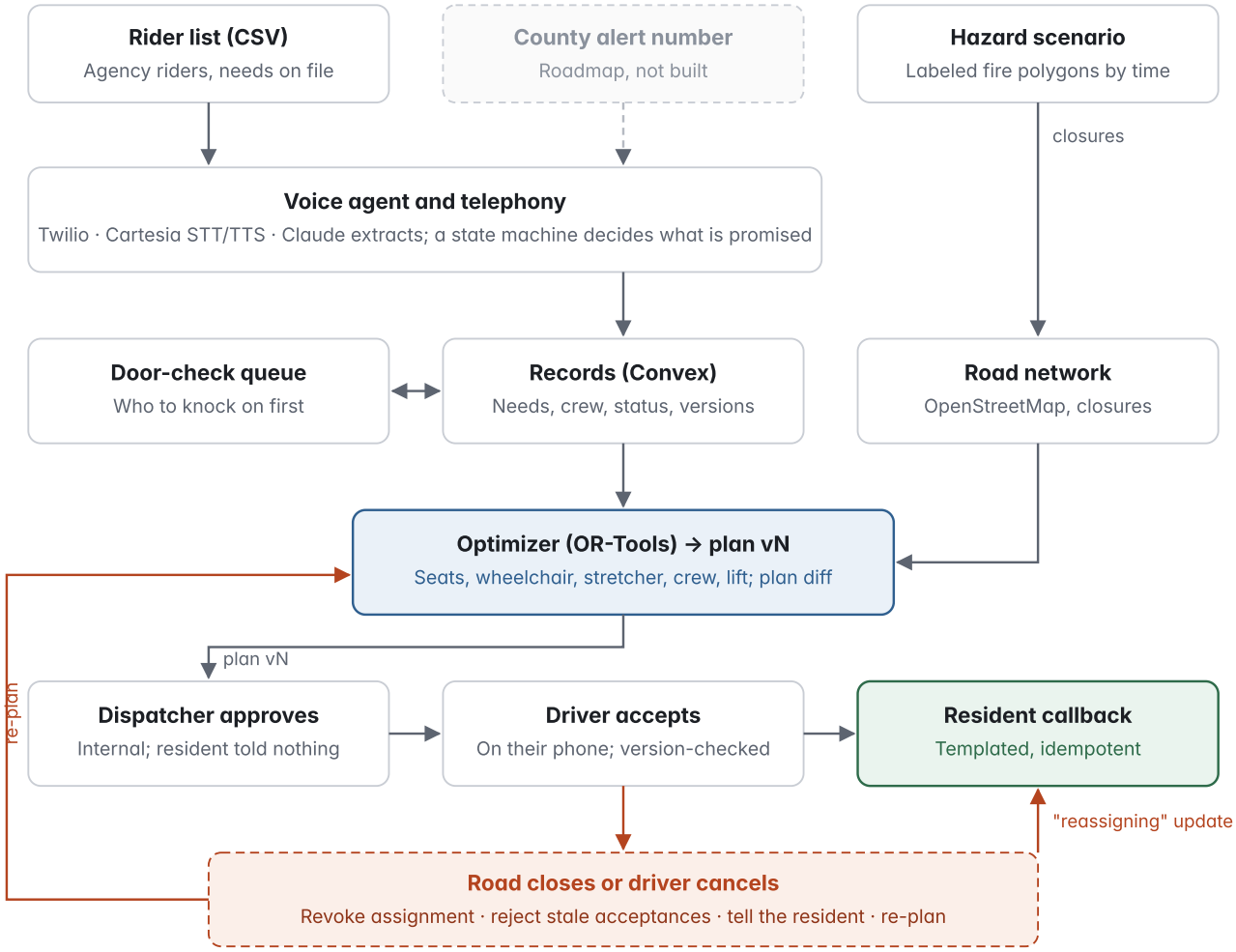
Technical architecture

One real-time database (Convex) that every component reads and writes. The depth is in four places: a voice agent that recovers from corrections, a dispatch optimizer with real compatibility constraints, versioned plans that stay correct when events conflict, and a simulator that measures honestly.

> **THE HARD PROBLEM, IN ONE SENTENCE FOR JUDGES**
>
> "This is dynamic vehicle routing where the deadlines move with the fire, you don't know who needs a ride until you reach them, and every promise has to stay true while the plan changes underneath it." Say it early in every pitch.

Depth rule: every component has one number that survives "how did you set that?" If the honest answer is "we hardcoded it," simplify and say so.

Every call becomes a request; nothing reaches the resident until a driver accepts



System architecture. Gray dashed: roadmap. Red: the change path, which is the center of the demo.

1. Calls: voice that recovers

- **Telephony.** Twilio Programmable Voice; each call streams 8 kHz audio over a WebSocket (Media Streams) to our call server, one async session per call. Upgrade off the trial tier and request a higher calls-per-second limit.
- **Speech loop.** Streaming speech-to-text (Cartesia's own if it holds up on 8 kHz phone audio, so Cartesia carries both directions), then Claude, then Cartesia for the voice. Voice activity detection handles turn-taking and interruptions.
- **Who decides what.** Claude interprets messy speech and phrases replies. A deterministic state machine decides what to ask next and what may be promised. Do not weaken that split to make Claude look more autonomous; it is the answer to Anthropic's "where could it go wrong."

- **Corrections as a feature.** Every field in the record has a source utterance and a confidence. A correction overwrites the field, logs the change, and re-runs compatibility for any pending assignment. The read-back before hanging up ("two people, one wheelchair, needs two people to lift, 100 Terrace St, is that right?") is where most corrections land.
- **Languages.** English and Spanish, end to end (recognition, extraction, voice, templates). Don't claim "any language."
- **Latency budget:** under 1 second per turn. Haiku 5.5 for turns; stream Claude's text to Cartesia sentence by sentence; record writes are fire-and-forget; a short backchannel ("okay, got it") covers tool calls. Show p50 and p95 live. Correct recovery matters more than shaving milliseconds.
- **Outbound dialer.** Priority queue by deadline and known need, drained by a token-bucket rate limiter sized to our calls-per-second limit. Answering-machine detection; voicemail leaves a short message and schedules a retry.
- **Text fallback.** If voice fails twice: a text with numbered replies plus free text Claude parses. If SMS registration hasn't cleared by Sunday, show it on a simulated phone panel labeled as such.

2. Records, plans and promises (Convex)

Tables: households, calls, requests, vehicles, shelters, road edges, **plans** (versioned), **assignments**, events, **statements** (every sentence that promised anything, linked to the status at that moment), nominations.

- **New household fields:** `assistance_required` (none, one-person assist, two-person lift), field-level source and confidence, correction history.
- **New vehicle fields:** `crew_size`, `lift_capable`, plus capacity per dimension (seats, wheelchair spaces, stretcher).
- **Plans are versioned.** Each plan has a version, a status (`proposed`, `approved`, `superseded`), and a diff against the previous version with a reason per change.
- **Acceptance is a version check.** An assignment stores the plan version it came from and a status (`offered`, `accepted`, `declined`, `expired`, `revoked`). The accept mutation succeeds only if that version is still current and the offer is still open. Convex mutations are transactional, so the check and the write happen atomically: a driver tapping "Accept" on plan v4 after v5 exists gets "This assignment changed," and no callback fires.
- **Offers expire.** Two minutes without a response marks the offer `expired` and re-offers.
- **Callbacks are idempotent.** Each callback is a scheduled function keyed by request, assignment and status. A repeated event is a no-op, so nobody gets a second, conflicting promise.
- **Revocation is a workflow.** Closure or cancellation sets the assignment to `revoked` and the request to `reassigning`, schedules the approved script immediately, and triggers a re-solve.
- **Also in Convex:** the reach ladder as a durable workflow (survives restarts, each household's step visible on the map); solver runs as actions that call the Python service and write back a

new plan version; an append-only audit log of approvals, offers, acceptances and statements, which the honesty eval reads.

Convex demo beat: show the stale acceptance being rejected, the audit trail of v4 to v5, and a duplicate event producing no second call. That is visibly "correct under conflicting events," which is more convincing than listing features.

3. Hazard input

- **Core:** labeled scenario polygons by time over the real west Altadena road graph, drawn from the published progression timeline. An edge closes when a polygon reaches it within a safety margin. On screen: "Scenario input, not a forecast."
- **After the gate:** the simple spread model returns as a scenario generator (wind-driven elliptical spread with fast marching via scikit-fmm on a uniform fuel grid). It makes closures emerge from simulated spread instead of a script. Claim "conservative within our assumptions," never "validated against satellite data": MODIS and VIIRS detections are about 1 km and 375 m, intermittent, and blocked by smoke.
- **It never marks anyone safe.** An unanswered household is always unknown. Say this unprompted.

4. Road network

- OpenStreetMap via OSMnx: edges with length, speed limit and lanes; free-flow travel times.
- Closures from the hazard input and from drivers' reports ("Lincoln Ave blocked"), plus residents' mentions on calls, extracted by Claude and geocoded, flagged for confirmation.
- Shortest paths from every vehicle, request and shelter node (Dijkstra on about 10,000 nodes in milliseconds); recompute only paths through changed edges.

5. Dispatch optimizer

A dynamic dial-a-ride problem: a mixed fleet, pickups and drop-offs, deadlines, re-solved whenever the world changes.

- **Requests:** pickup node, eligible shelters (any one), load vector, assistance required, boarding time (about 2 minutes walking, 5 walker, 10 wheelchair, 20 bedbound or two-person lift), deadline, priority weight.
- **Vehicles:** position, passengers on board, capacity vector, crew size, lift capability, shift end.
- **Compatibility is a hard constraint, and it is visible.** A request needing a two-person lift can't go to a one-driver van, even with a free wheelchair space. The rescue map shows rejected options with their reason ("1 wheelchair space, needs 2"; "crew of 1, needs 2").

$$w_r = \nu_r \,/\, \max(\text{slack}_r, \varepsilon), \quad \text{slack}_r = L_r - \text{earliest arrival}_r$$
$$\text{maximize} \;\; \Sigma_r \, w_r \, x_r - \lambda \, \Sigma_r \, \text{late}_r - \mu \, \Sigma_k \, \text{time}_k - \nu \cdot \text{reassignments}$$

- **Constraints:** pickup before drop-off on the same vehicle; capacity per dimension; crew and lift compatibility; time windows; maximum ride time for oxygen users; shelter capacity.
- **Solver:** OR-Tools Routing with pickup-and-delivery pairs, eligible shelters as a disjunction, three capacity dimensions plus a time dimension, compatibility as allowed-vehicle sets, each request optional with a drop penalty equal to its weight. Parallel cheapest insertion, then guided local search, 3-second limit.
- **Re-planning:** event-driven, debounced to every 5 to 10 seconds, warm-started from the current plan, each vehicle's next stop frozen unless it is invalidated.
- **Escalation:** requests the solver can't serve in time go to the top of the dispatcher's list for fire and EMS.
- **Fallback:** the competent heuristic (below) is also the fallback dispatcher if the solver service fails.

> **"DIDN'T YOU JUST CALL OR-TOOLS?"**
>
> "OR-Tools is the search engine; the problem is ours. We built the dynamic formulation with deadlines from the hazard, the compatibility rules including crew, warm-started re-solves that don't jerk drivers around, versioned plans that stay correct when an old acceptance arrives late, and the baseline we beat."

6. Simulator and evaluation

Discrete-event simulation (SimPy) of a night: the hazard advances, calls are answered or missed, vehicles drive, roads close, drivers sometimes cancel. Every assumption is labeled.

- **Three policies on identical scenarios:** today's process (911 first come, first served, nearest unit, radio delays; context only); a **competent dispatcher heuristic** (earliest deadline first, type and crew matched, greedy insertion), which is the baseline we must beat; and ours (OR-Tools, warm-started re-planning).
- **Three separate results:**
  1. **Dispatch only:** both dispatchers receive identical requests at identical times. This is the honest optimizer number.
  2. **Outreach only:** the heuristic dispatcher, but with requests arriving when our calls would surface them.
  3. **Combined.**
- **Metric:** additional simulated pickups completed before the scenario deadline. Also worst wait, wait for wheelchair and bedbound riders, and vehicle-minutes. About 100 seeds, reported with ranges. Never "lives saved."
- **Expect a modest dispatch gap.** A competent heuristic is strong; single digits to low double digits is a good result. Script it now: "A good dispatcher with a whiteboard does fine when it's quiet. The gap opens when [the fleet is tight / closures stack up]; here's by how much."

- **Sensitivity panel:** fleet size, answer rate, hazard speed. One honest line about where we don't help (for example, below a certain fleet size nothing reaches the last households in time).
- **Ground-truth households:** from the synthetic population (section 8). State the limitation openly: the simulator evaluates a policy; it doesn't validate predictions about real households.

Call quality, reported per field

1. **Persona eval (scale):** about 200 text personas with ground-truth cards (hard of hearing, partial address, panic, Spanish, tangents, a second voice, **mid-call corrections**). Accuracy reported separately for address, passenger count, mobility, and assistance required; correction success rate; completion time.
2. **Audio eval:** about 30 synthetic-voice callers from a different TTS provider (so we don't test Cartesia against itself) calling our real Twilio number, with noise and slow speech. Report the drop from text.
3. **Usability tests with people:** calls with older relatives playing persona cards, presented honestly as usability testing, not as evidence about riders or dispatchers.

7. Door-check queue (David)

Calls scale; door-knock teams don't. With 6 teams and 400 unreached homes, order matters. The queue is a transparent score, not a black-box posterior:

$$\text{priority}_h = (p_h \cdot u_h) / \tau_h + \alpha \cdot \text{wait}_h$$

- p_h: need estimate. Known need from the rider list or a call; a neighbor nomination raises it sharply; otherwise the block-group base rate (ACS: 65+, disability, no vehicle, 65+ living alone). **Spatial smoothing** from answered neighbors is one term we can switch on and off on screen.
- u_h: urgency from the hazard deadline. τ_h: travel time for the nearest team.
- **wait$_h$**: minutes on the list. This aging term guarantees no household, however isolated, stays last forever.
- **Fairness tests in the simulator:** 30% of nominations wrong; one neighborhood answering at half the rate; isolated homes with no answering neighbors. Report **worst-case** time to check, not only the average, against random and nearest-first orderings.
- **Guardrails:** the score only orders visits; nobody is skipped because of it; estimates shown as estimates; no individual risk scores on screen.

8. Synthetic population (David)

- **Method (standard, as in PopGen and ActivitySim):** draw real ACS PUMS household records from the PUMA(s) covering Altadena, then reweight them with iterative proportional fitting to block-group totals (65+, disability, no vehicle, living alone). Because whole real records are resampled, the dangerous combinations (75+, ambulatory difficulty, alone, no car) are kept.

- **Why not flow matching:** public PUMS is located only to the PUMA (about 100,000 people), so records can't be conditioned on their own block group. Resampling plus IPF already preserves the joint structure. A demography judge would see a learned model here as not beating the standard method.
- **Placement:** onto real building footprints, clustered around real senior housing and care facilities from OpenStreetMap.
- **Feeds:** simulator ground truth, the 200 persona cards, and the door-check queue's base rates.

After the 6pm gate: extras, in order

Nothing here starts until the core flow works end to end. Then the gate, not this list, decides what gets added.

1. **Simulator results** with the dispatch, outreach and combined split, and the sensitivity panel.
2. **Audio eval and the full red-team set,** including the 10 change cases and the implied-reassurance grader.
3. **Door-check fairness tests.**
4. **Simple spread model** as a scenario generator (section 3). *Film the Devpost video before starting this.*
5. **Congestion and departure timing, as labeled exploration** on the under-the-hood screen: "What happens if everyone leaves at once?"
   - Edge travel time with the BPR curve: $t = t_0\,(1 + 0.15\,(v/c)^4)$, with residents who drive loaded onto the network by departure time.
   - Departures from a drift-diffusion model, $dX = \mu(t)\,dt + \sigma\,dW$, leaving when X crosses a threshold; two classes (evacuation-keen and evacuation-reluctant, per Berkeley's survey of California evacuees), shape fit to published departure curves.
   - Framing: the effect of a phone call on departure is an **assumption explored** in simulation, not a finding. Skip the time-expanded min-cost flow and the coupling loop unless everything else is done.

Out of scope this weekend

ITEM	WHY IT STAYS OUT, EVEN WITH TIME
Flow matching and the spline flow	The training join doesn't work with public data, and resampling real records already solves the problem it was meant to solve. Easier to attack than to defend.
Volunteer tier	Invites liability, perimeter and trust questions and pulls focus from confirmed professional pickup.
FIRMS validation number	The data can't support address-level arrival-time error.

ITEM	WHY IT STAYS OUT, EVEN WITH TIME
Federated-learning close	Out of the script. Kept as a Q&A answer about multi-agency scale.
"Any language," Himalayas roadmap line	Unprovable this weekend and a change of subject in the close.

Where to push for maximum impressiveness

1. **The change moment:** road closes after acceptance, stale acceptance rejected, resident told within seconds, new driver found.
2. **The compatibility moment:** a nearby van rejected for space or crew, with the reason on screen.
3. **The correction moment:** a caller changes a consequential answer and the assignment changes with it.
4. **Honesty numbers:** template promises, implied reassurance from the grader, and revocation-to-notification time.
5. **Dispatch-only result** against the competent heuristic, with ranges and the line about where we don't help.
6. **Call accuracy per field,** across personas, audio, and usability tests.
7. **Measured latency,** p50 and p95.
8. **Door-check fairness:** worst-case time to check under wrong nominations and uneven answer rates.

Make the computation visible. Every re-plan shows requests, vehicles, solve time, plan version and reassignments. Every simulator number shows a range.

Team roles

Four owners, one piece each, meeting at the Convex schema.

ROLE	OWNER	BUILDS	HANDS OFF TO
Voice and calls	Jerry	Telephony, the voice loop, the state machine with corrections, Spanish end to end, honesty states and templates (including `reassigning`), the reach ladder workflow, idempotent callbacks, text fallback, latency budget, the red-team set, the audio eval	Records: households and statements
Optimizer, plans and simulator	Boris	Road graph and closures, OR-Tools with crew and lift constraints, plan versions and diffs with reasons, stale-acceptance rejection and revocation (callbacks with Jerry), the competent heuristic (also the fallback), the simulator with the three-way split and sensitivity panel; after the gate, congestion and departure exploration	Rescue map: plans, diffs, metrics

ROLE	OWNER	BUILDS	HANDS OFF TO
Modeling and evaluation	David	Door-check queue and fairness tests, synthetic population (PUMS plus IPF), the 200-persona eval harness with corrections and per-field scoring, the implied-reassurance grader, hazard scenario polygons; after the gate, the simple spread model	Field teams: door order. Simulator: households. Optimizer: deadlines
Rescue map, driver app, demo and evidence	Brendan	The rescue map with plan versions, rejected-option reasons and one-click approval; the driver acceptance page on a second phone; Rider Check CSV import and zone drawing; the under-the-hood screen; practitioner outreach and the "what we changed" slide; usability tests; the pitch (2- and 3-minute cuts); the Devpost video; Notability notes	Judges

Timeline

v8 is adopted Saturday morning. The core flow must work end to end by Saturday 6pm; everything after is extras, evidence and polish.

Fri 6pm to Sat 9am

Lock the plan and build the four pieces (v7)
Voice loop, records, road graph, rescue map shell, outreach sent.

Sat 9am to 12pm

Re-scope to v8 and test with people
Freeze cut items. Add crew, lift and version fields. Usability calls with older relatives. Chase one practitioner conversation; ask mentors with public-safety backgrounds for 10 minutes.

Sat 12pm to 6pm

Integrate the Rider Check flow end to end
CSV, call, correction, compatible assignment, approval, driver acceptance on a second phone, callback, closure, stale rejection, reassigning update, new acceptance.

Sat 6pm

Gate: the core flow works end to end

Sat 6pm to Sun 2am

Extras in order, add-ons in parallel
Simulator split, red-team and audio eval, fairness tests. Film the video. Then the spread model and congestion exploration. Add-ons: one owner, one-hour timebox each.

Sun 2am to 8:30am

Freeze, polish, submit
Feature freeze at 4am. Edit the video, write Devpost and READMEs, rehearse both cuts and the Q&A, opt into every track.

Sun 8:50am — **Submitted on Devpost**

Sun 10:30am — **Judging at our table**
Closing ceremony at 2pm.

The 6pm gate includes: the full Rider Check flow above; honesty states including `reassigning`; corrections reaching the assignment; plan versioning with stale-acceptance rejection; the competent heuristic running as fallback; measured latency. If the core isn't working at 6pm, drop every extra and spend the night on reliability and backups.

Do now (carry-overs and v8 changes)

- **Jerry:** confirm Twilio is off trial with a higher calls-per-second limit and SMS registration started; test Cartesia speech-to-text on 8 kHz audio; pick the Spanish voice; add correction handling and the read-back to the state machine.
- **Boris:** add `crew_size`, `lift_capable` and `assistance_required` to the solver; add plan versions and the version-checked accept mutation; make the heuristic the fallback.
- **David:** switch the population to PUMS plus IPF; build the door-check score with the smoothing toggle; draw scenario polygons from the published progression timeline; add corrections to the persona cards.
- **Brendan:** build the driver acceptance page; fix the narration ("order," not "warning"; counterfactual framing); follow up on outreach with one question: "What would change how you'd use this?"; book public-safety mentors.
- **Team:** remove cut items from the task list; pick the product name.

Customers, users and the path to production

The goal is no longer collecting quotes. It is one sentence: "We asked X. A dispatcher told us Y. We changed Z."

- **Customer:** paratransit agencies like Access Services first; county emergency management later.
- **Users:** riders who can't leave on their own, the agency's dispatchers, and its drivers.
- **Practitioner conversation (highest priority):** an Access Services dispatcher or manager, a paratransit driver, or an emergency manager. Even one real conversation moves Anthropic, Microsoft and People-First more than any model. Ask: how would an assignment reach a driver today? Who is allowed to approve? What would make you distrust this?
- **Advocates:** Disability Rights California, an LA independent living center, LA County CERT. One question each: what would make you trust or distrust an AI agent calling people during an evacuation? Their answer shapes the "if you can leave safely" script.

- **Usability tests:** 5 to 10 older adults with persona cards, recorded with permission. Ask what confused them, whether they trusted it, whether they'd have left. Fix and re-test one person.
- **Approach Altadena residents only with care;** fire survivors don't owe us their time.
- **Report exactly what we got,** for example "1 dispatcher call, 2 advocate replies, 8 usability tests with adults 70+." Real but small beats impressive but inflated.

Path to production (for Microsoft)

1. **Supervised drill** with one agency, fictional riders, its own dispatchers at the controls.
2. **Shadow mode** during real warnings: the system builds plans alongside the agency's dispatch, and we compare, without calling anyone.
3. **Pilot** on the agency's real rider list under its authority. County mode only after this works.

THE QUESTION A FORWARD-DEPLOYED ENGINEER WILL ASK	OUR ANSWER
Who authorizes the calls and assignments?	The agency. The rescue map is their dispatcher's screen; nothing goes out without their approval.
Who handles urgent escalations?	The agency's dispatcher, with red records at the top of the list and a handoff to 911.
How do assignments reach drivers?	A link on the driver's phone this weekend; their existing driver tablets later.
What if a rider needs more help than a vehicle can give?	Compatibility blocks it; if nothing compatible exists in time, it escalates to fire or EMS.
What happens when calls, location feeds or the solver fail?	Text fallback; the heuristic dispatcher; an exportable call list; the audit log shows what was said.
Integrations	CSV or rider-database export, the agency's vehicle GPS feed, driver acceptance, escalation to 911.

Responsible design

- Rider data is protected: fictional riders in every demo; real data only under the agency's authority; deleted when the incident closes; role-based access (Auth0).
- No individual risk scores shown; a human approves every plan; the honesty rules; the audit log.
- **AI voice calls:** the FCC ruled in 2024 that AI-generated voices count as "artificial voice" under the TCPA. Our answer: the agency calls its own riders for an emergency purpose, which falls under the TCPA's emergency exception; agency counsel would confirm before any real deployment.

Demo plan

One rider whose situation changes, handled correctly the whole way through, in about 3 minutes in a loud room.

Table setup

- Big screen: the rescue map over the real west Altadena street grid, with the hazard scenario labeled "scenario input."
- A sample rider CSV of fictional riders, ready to import, and a zone to draw.
- 2 or 3 "rider phones" with persona cards that include a correction, for example: "Carmen, 79, speaks Spanish. You first say it's just you. Then remember your son is home: he uses a wheelchair and needs two people to lift him."
- **A driver phone** held by a teammate, showing the offer and an "Accept" button. This is what makes acceptance a moment, not a database field.
- A solver panel that updates on every re-plan: "Plan v5 · 12 vehicles · 46 requests · 1.8s · 2 reassignments · 1 stale acceptance rejected."
- Under-the-hood screen: architecture, latency p50 and p95, per-field accuracy, honesty numbers, simulator results with ranges, and the honesty panel (each request's status beside the last thing the agent said).

The 3-minute script

1. **0:00 Story.** A father and son. The order came at 3:25am with no warning first. Vans were ready at 4am and nobody called. Three 911 calls, starting at 6:03. "We can't say what would have saved them. We can build the connection that was missing."
2. **0:20 Counterfactual setup.** "What if the paratransit agency started calling its riders the moment a warning covered their street?" Import the fictional rider CSV, draw the zone, calls start.
3. **0:35 The call.** The judge answers as Carmen, in Spanish or English. Midway, they correct themselves: the son is home, wheelchair, two-person lift. The record changes on screen. They ask "is someone coming?" The agent: "Not yet. You're on the list as high priority, and I'll call back the moment a driver accepts."
4. **1:05 Technical line and compatibility.** "Under the hood, this is vehicle routing where deadlines move with the fire and the promises have to stay true while the plan changes." The nearest van is rejected on screen ("crew of 1, needs 2"); a two-person crew is assigned. The dispatcher approves plan v4. "Approval isn't a promise, so Carmen hears nothing yet."
5. **1:30 Acceptance.** The teammate taps Accept on the driver phone. The judge's phone rings: "A wheelchair-accessible van with a two-person crew is 6 minutes away. The driver's name is Maria."
6. **1:50 The change.** The scenario closes the van's road. Plan v5 re-solves. The driver's late tap on the v4 assignment is rejected. Carmen's phone rings: "Your driver can no longer reach you.

You're still at the top of the list." A new driver accepts; a new callback.

7. **2:20 Proof.** Dispatch-only gain over a competent heuristic on identical requests, with ranges; per-field call accuracy; honesty results including revocation cases; what a practitioner told us and what we changed. Labeled as simulation where it is.
8. **2:45 Close.** "Ready for a supervised drill on fictional riders, then shadow mode beside an agency's dispatchers. Wildfire is the first hazard; floods and hurricanes need their own rules, on the same pipeline." Last line: "The vans were ready at 4am. The call came at 6:03. We connect them."

The 2-minute cut (Cartesia's format and quick-visit judges): demo first, story last

1. **0:00** The phone rings; the judge plays Carmen, including the correction and the honest "not yet" (about 35 seconds).
2. **0:35** Van rejected for crew; approval; the driver accepts; callback.
3. **1:05** Road closes; stale acceptance rejected; "reassigning" call; new driver.
4. **1:35** One line of story; what Cartesia does in the pipeline (both directions, if STT holds), latency, per-field accuracy, and known limitations.
5. **1:55** "Ready for a supervised drill on a paratransit rider list."

Devpost video (3 minutes max). Open on a dark room and a ringing landline; cut to the rescue map. Then the same single-rider flow. Film Saturday night, before the spread model and congestion work.

Q&A prep

LIKELY QUESTION	ANSWER
"The driver accepts plan v4 after a road closure invalidated it. What happens?"	Show it: the accept is rejected atomically, no callback fires, the audit log shows v4 superseded by v5.
"What if the person isn't sure, or changes their answer?"	Corrections overwrite the field, re-run compatibility, and are measured (correction success rate in the persona eval).
"Would this have saved the father and son?"	"We can't say that. The reporting shows vans ready and nobody called. We built that missing connection; whether it saves a given life depends on things we don't control."
"Is your fire model validated?"	"It's a scenario input. Satellite detections can't validate arrival times at individual homes, so we don't claim that."
"Don't isolated households always end up last?"	The waiting-time term ages every household up the list; show the worst-case result from the fairness test.
"Isn't the simulator grading its own homework?"	Ground truth comes from real census records reweighted to real totals; it evaluates a policy, not a prediction, and we say so on screen.

LIKELY QUESTION	ANSWER
"How does this scale to other agencies?"	Each agency runs its own instance on its own riders. Across agencies, a shared need model could improve through cross-agency (federated) training without raw data leaving any agency. Not built.
"Isn't AI calling people regulated?"	The TCPA answer above.
"Why voice instead of an app?"	No install during an evacuation; landlines and flip phones work; riders who can't use screens can still answer.

Prize tracks

Fewer tracks, each with work a judge can see. Every sponsor penalizes bolt-ons.

TRACK	PRIZE	WHAT THE JUDGE MUST SEE	TIER
Cal Hacks: Overall, Most Technically Complex, People-First	MacBook Air down to Apple Watch	People-First is our clearest fit. Most Technically Complex comes from dispatch staying correct under changing constraints, shown live on the under-the-hood screen, not from the number of models.	CORE
Cartesia	Meta glasses (TBD)	Voice is core: no app to install. One complete flow including a correction and an interruption; per-field accuracy; latency; Spanish. The 2-minute demo-first cut and a README with Cartesia features used and known limitations. They favor a finished prototype over a big idea; so do we now.	CORE
Convex	$5,000	Correctness under conflicting events: version-checked acceptance, idempotent callbacks, revocation as a workflow, the reach ladder, the audit log. Show the stale acceptance rejected live.	CORE
Anthropic	Not posted yet	Named community: paratransit riders in LA County wildfire zones. A real problem: what a practitioner told us. It works: one flow done well. Claude interprets messy, corrected speech in two languages; code controls every commitment, and we show where Claude could go wrong (implied reassurance, measured).	CORE
Microsoft: Built for the Real World, Customer Impact	Surface each	The question-and-answer table above, the three-phase path, the dispatch-only result, responsible design, and the full flow end to end. Strongest if a practitioner conversation happened.	CORE

TRACK	PRIZE	WHAT THE JUDGE MUST SEE	TIER
Apify: Best Actor	$400	A public "California evacuation status" Actor that normalizes county and state evacuation zones and road closures, and triggers Rider Check when a zone with riders goes to warning. Useful to others on its own. Top-5 overall is a long shot; aim for Best Actor.	AFTER GATE
MLH: Auth0	Headphones each	Dispatcher login, role-based access to the at-risk list, step-up authentication before approving a plan.	AFTER GATE
Notability	1 year Pro	Only if notes are already being kept: sketches, mentor feedback as audio, decisions as they happened. Added after the fact scores zero.	AFTER GATE
Figma	Verify at the event	The live prize feed now lists gift cards and a recruiting session, not the $2,000 prizes in v7. Keep the stage log only if it's already running.	OPT-IN
Planet-First, Hacker's Choice, MLH opt-ins (domain, Gen AI, hosting, Commit Fellowship)	Assorted	Opt in on Devpost; no extra build work.	OPT-IN

Cut, and why

TRACK	REASON
Fetch.ai	The whole primary workflow must run inside an ASI:One conversation, with a separate 3 to 5 minute video. A publicly discoverable agent that accepts "check on this address" from anyone is a flooding risk during a real emergency, which contradicts our privacy story.
Bitdeer	Model Studio would only host eval personas; 20% of the score is meaningful use.
SpaceXAI	"Real space data goes in." Without the FIRMS-driven fire model there is no basis.
Mana	30% of the score is "did Mana let you build something better"; one import screen doesn't show that. Prize amount not posted.
Meta	Judged on strengthening human connection; our core is dispatch.
Moss	Returning-caller memory and shelter lookup don't change the core flow enough to justify the integration.

Rules. Nothing below Core is touched before the 6pm gate. Each add-on has one owner and a one-hour timebox; whatever isn't working by then doesn't go in the demo. Every sponsor tool gets a README line saying where it runs.

Risks and backups

RISK	BACKUP
Scope creep, including from the "we can pull it off" list	The gate decides. Extras only after 6pm, in the stated order.
The agent promises help that isn't coming	Statuses in code; ETAs only by template; the implied-reassurance grader; red-team set run before the demo
A promise becomes false after acceptance	`reassigning` status, revocation workflow, version-checked acceptance, measured time to notify
A correction doesn't reach the assignment	Correction cases in the persona eval; compatibility re-runs on every field change
No practitioner conversation by Sunday	Mentors with public-safety backgrounds at the event; report exactly what we got; don't present usability tests as customer evidence
Dispatch gain over the heuristic is small	Script the honest version; show where the gap opens; lead with correctness, not margin
Cartesia STT fails on 8 kHz audio	Another streaming STT; Cartesia stays the voice; say so in the README
Texts blocked because SMS registration hasn't cleared	Voice is the main path; a simulated phone panel for the text fallback, labeled
Twilio trial limits or spam flagging	Upgraded account; demo on team-owned phones
Loud room	Phone to the ear; live transcript on screen
Voice latency over 1 second	Haiku turns, sentence streaming, fire-and-forget writes, backchannel
Voice or venue wifi fails live	Phone hotspot; recorded calls; the pre-recorded video
Overclaiming the case study or the impact	Counterfactual framing; "additional simulated pickups," never lives saved; every simulated number labeled
"Isn't AI calling people regulated?"	The TCPA emergency-purpose answer
Exposing the at-risk list	Fictional riders only; role-based access; deletion on incident close; no public intake endpoint

Rules we keep: only authorized agencies send emergency alerts; a human approves every plan; nothing reaches a resident before a driver accepts; the list of vulnerable residents is sensitive data.

Open decisions

- Product name
- Whether Cartesia's speech-to-text holds up on 8 kHz phone audio

- Which neighbor-question wording and "can't leave" wording residents and advocates find natural
- Who plays the driver at the table (needs to be free for the full judging window)

Sources

- LAist: Despite three 911 calls, two homebound disabled men died in the Eaton Fire waiting for rescue
- KQED: Two homebound disabled men died in the Eaton Fire
- Firehouse / LA Times: UL FSRI findings on the Palisades and Eaton fires
- LA County: Eaton Fire after-action synopsis (McChrystal Group)
- LAist: LA County looks to improve support for vulnerable, older people
- KFF Health News: Disability advocates warn against registry false assurances
- Disability Rights Texas: Let's talk about STEAR
- UN News: Needs of persons with disabilities ignored during disasters
- NBC News: Half of Katrina victims were 75 or older
- Firehouse: Elderly particularly vulnerable during wildfires (Camp Fire)
- Scripps News: Lahaina victims identified
- IDMC: Global Report on Internal Displacement 2026 (summary); NRC press release, May 2026
- U.S. Census Bureau: ACS Public Use Microdata Sample documentation (PUMA geography)
- NASA Earthdata: MODIS and VIIRS active fire detection resolution and limitations
- FCC: Declaratory ruling on AI-generated voices under the TCPA (2024)
- Cal Hacks 13.0 live prize feed (hive.hackberkeley.org), checked Oct 24, 2026
- UC Berkeley ITS: Understanding California Wildfire Evacuee Behavior and Joint Choice Making
- UC Berkeley TSRC: Review of California Wildfire Evacuations from 2017 to 2019
- Outside reviews of v7 (Oct 23 to 24): Claude judge review, CalHacks 13 strategy review, ChatGPT review